PROJECT WORK
OF
BASIC MATHEMATICS

CLASS -12

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CHAPTER:1

Permutation & Combination

a) Set of objects all different p(n,r) = nP_r

b) Permutation of objects not all different

→Total number of permutations =

$$\frac{n!}{p! \times q! \times r!}$$

c) The total number of permutation of a set of n objects arranged in a circle = (n-1)!

1) Combination C(n,r) $= \frac{n!}{(n-r)!}$
2) C (n_1 r) =C (n_1 n-r)
3) r + r = n
4) C (n_1 r) + C (n_1 r-1) = 0(n+1, r)

Chapter – 2

<u>Binomial Theorem</u>

1. **For any positive integer n,**

$$(a+x)^n = C(n,0)a^n x^0 + C(n,1)a^{n-1}x + C(n,2)a^{n-2}x^2 + \cdots + C(n,r)a^{n-r}x^r + \cdots + C(n,n)x^n$$

2. **<u>General Term</u>**

$$(a+x)^n = \sum_{r=0}^{n} C(n,r)\, a^{n-r} x^r$$

$$(a-x)^n = (-1)^r C(n,r) a^{n-r} x^r$$

3. **<u>Binomial Coefficients</u>**

The coefficients $C(n,0), C(n,1), \ldots, C(n,n)$ in the expansion of $(a+x)^n$ are known as binomial coefficients.

1) $2^n = C(n,0) + C(n,1) + C(n,2) + \cdots + C(n,n)$
(Sum of all binomial coefficients)

$$2)\ C_0 + C_2 + C_4 + \cdots = C_1 + C_3 + C_5 + \cdots$$

(Sum of odd = Sum of even)

$$nCr \equiv C(n,r) \equiv \frac{n!}{(n-r)!\ r!}$$

Chapter – 3

Complex Number

1. The complex number ; $z = x + iy$may be written in

$$z = r(\cos\ \theta + i\sin\ \theta)\text{[trigonometric form]}$$

where $r = \sqrt{x^2 + y^2}$, $\tan\ \theta = \frac{y}{x}$

$r =| z |$= modulus of z, θ= amplitude or argument of z

2) **Euler's Formula**

$$e^{i\theta} = \cos\ \theta + i\sin\ \theta$$

3) If $z_1 = r_1(\cos\ \theta_1 + i\sin\ \theta_1)$& $z_2 = r_2(\cos\ \theta_2 + i\sin\ \theta_2)$

$$z_1 z_2 = r_1 r_2\left[\cos\ (\theta_1 + \theta_2) + i\sin\ (\theta_1 + \theta_2)\right]$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\left[\cos\ (\theta_1 - \theta_2) + i\sin\ (\theta_1 - \theta_2)\right]$$

4) **De Moivre's Theorem**

If n is any positive integer,

$$\left[r(\cos\ \theta + i\sin\ \theta)\right]^n = r^n(\cos\ n\theta + i\sin\ n\theta)$$

The cube roots of unity. (2 marks)

1) Each imaginary cube root of unity is the square of the two other.

$$\omega = \omega^2$$

2) The product of the two imaginary cube root of unity is equal to 1.

$$\omega \cdot \omega^2 = 1$$

$$\omega^3 = 1$$

3) The sum of the three cube roots of unity is zero →

$$[\ 1 + \omega + \omega^2 = 0\ ]$$

CHAPTER-4

Sequence and series

Formula:

1. Sum of the first n natural number

$$S_n = \frac{n(n+1)}{2}$$

2. Sum of the first n even natural number

$$S_n = n(n+1)$$

3. Sum of the first n odd natural number

$$S_n = n^2$$

4. Sum of the square of the first n natural number

$$S_n = \frac{n(n+1)(2n+1)}{6}$$

5. Sum of the cube of the first n natural number

$$S_n = \left(\frac{n(n+1)}{2}\right)^2$$

6.

$$t_n = a + (n-1)d$$
$$S_n = \frac{n}{2}[2a + (n-1)d]$$

7.

$$t_n = ar^{,n-1}$$
$$S_n = \frac{a(r^{,n} - 1)}{r - 1}$$

Chapter-5

MATRIX AND DETERMINANTS

1) Cramer's Rule

Rules: → find 'D' Determinants

→ find 'Dx' determinants of x

→ find 'Dy' determinants of y

2) Row Equivalent Matrix

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Row = 2

Column = 2

C_1 C_2

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Row = 3

Column = 3

C_1 C_2 C_3

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \qquad \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

3) Inverse matrix method...

CHAPTER -6

PROPERTIES OF TRAINGLE

1. The Cosine Law

In any ΔABC,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \text{ i.e.,} a^2$$
$$= b^2 + c^2 - 2bc\cos A$$
$$\cos B = \frac{c^2 + a^2 - b^2}{2ca} \text{ i.e.,} b^2$$
$$= c^2 + a^2 - 2ca\cos B$$
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} \text{ i.e.,} c^2$$
$$= a^2 + b^2 - 2ab\cos C$$

2) The Sine Law

In any ΔABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

where R is the radius of the circum-circle.

3) Projection Laws

In any ΔABC,

$$a = c\cos B + b\cos C$$
$$b = c\cos A + a\cos C$$
$$c = a\cos B + b\cos A$$

4. The Half Angle Formula

In any ΔABC, let $s = \frac{a+b+c}{2}$

a.

$$\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

b.

$$\sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$$

c.

$$\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$$

5. The Area of a Triangle

In any ΔABC,

a.

$$\Delta = \frac{abc}{4R}$$

b.

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

Chapter – 7

Solution of Triangle (3 marks)

Let a triangle have sides a, b, c opposite to angles A, B, C respectively.

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1. Sum of angles property:

$$\angle A + \angle B + \angle C = 180^\circ$$

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2. Law of Sines:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

---

3. Law of Cosines:

$$\cos\ A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos\ B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos\ C = \frac{a^2 + b^2 - c^2}{2ab}$$

Q.6 Parabola and Standard Equation

→ **A parabola is the locus of a point which is equidistant from a fixed point (called the focus) and a fixed line (called the directrix).**

(a) Standard Equation of a Parabola

Let AS = a. Then the coordinates of Z, A and S are respectively (−a, 0), (0, 0) and (a, 0) and the equation of the directrix is

x + a = 0

Let P(x, y) be any point on the parabola. Join PS and draw PM perpendicular to ZY.

Then

PS = PM

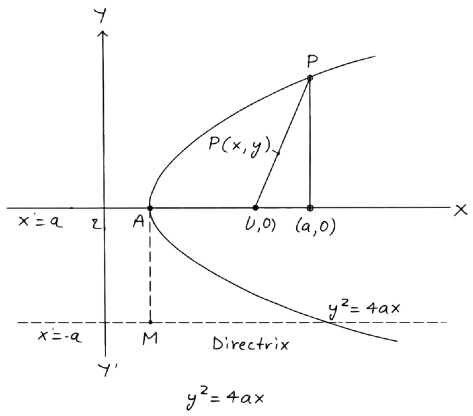
i.e.

$PS^2 = PM^2$

or $(x-a)^2 + y^2 = (x+a)^2$

or $y^2 = 4ax$

Note 2:



If **a is positive,**

$$y^2 = 4ax$$

is a parabola whose axis is the x-axis and the curve is entirely to the right side of the y-axis.

The parabola

$$y^2 = -4ax$$

lies to the left of the y-axis.

Table (As in the Image)

Equation of parabola	Vertex	Focus	Equation of the directrix	Axis	Length of latus rectum
$y^2 = 4ax$	(0,0)	(a,0)	x = −a	y = 0	4a
$x^2 = 4ay$	(0,0)	(0,a)	y = −a	x = 0	4a
$(y-k)^2 = 4a(x-h)$	(h,k)	(h + a, k)	x = h − a	y = k (parallel to x-axis)	4a
$(x-h)^2 = 4a(y-k)$	(h,k)	(h, k + a)	y = k − a	x = h (parallel to y-axis)	4a

Equation of the Tangent to the parabola

$y^2 = 4ax$at $(x_1, y_1)$is, (Point value)

$$xx_1 = 2a(y + y_1) \text{——①}$$

Condition of Tangency of a straight line to a parabola (Slope)

$$y = mx + \frac{a}{m}$$

Point of contact,

$$(x, y) = \left(\frac{a}{m^2}, \frac{2a}{m}\right)$$

Equation of Normal,

$$y - y_1 = -\frac{y_1}{2a}(x - x_1)$$

or

$$y = mx - 2am - am^3$$

Vector

Chapter – 9

Product of Vectors

Scalar product of two vectors

The scalar product of two vectors →a and →b denoted by →a · →b is defined by

$$\vec{a} \cdot \vec{b} = |\vec{a}| \; |\vec{b}| \cos\theta$$

Projection of →a on →b

Projection of $\vec{a}$ on $\vec{b}$

$$= \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

$$\vec{i} \cdot \vec{i} = 1$$
$$\vec{j} \cdot \vec{j} = 1$$
$$\vec{k} \cdot \vec{k} = 1$$

(If vectors are perpendicular, scalar product = 0)

cosθ → scalar product

Vector product of two vectors

a)

$$\vec{a} \times \vec{b} = |\vec{a}| \quad |\vec{b}| \sin\theta\, \hat{n}$$

where $\hat{n}$is unit vector normal to →a and →b

---

b) Unit vector

$$\text{Unit vector} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

---

c)

$$\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| \quad |\vec{b}|}$$

Note:

$$\vec{i} \times \vec{j} = \vec{k}$$
$$\vec{j} \times \vec{k} = \vec{i}$$
$$\vec{k} \times \vec{i} = \vec{j}$$

(If two vectors are parallel, sinθ = 0 → vector product = 0)

Chapter – 10

Correlation and Regression (5 marks)

Correlation (r):

$$r = \frac{N\sum xy - \sum x \sum y}{\sqrt{\left(N\sum x^2 - (\sum x)^2\right)\left(N\sum y^2 - (\sum y)^2\right)}}$$

Regression of y on x:

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

where

$$b_{yx} = \frac{N\sum xy - \sum x \sum y}{N\sum x^2 - (\sum x)^2}$$

Regression of x on y:

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

where

$$b_{xy} = \frac{N\sum xy - \sum x \sum y}{N\sum y^2 - (\sum y)^2}$$

Probability

FORMULAE AND IMPORTANT POINTS

1.

$$P(E) = p = \frac{\text{Favourable number of cases}}{\text{Total number of cases}} = \frac{m}{n}$$

2. **Properties of Probability**

If $p$denotes the probability of happening of an event, then the following properties of the probability are always true.

i. $p + q = 1$

ii. $P(\text{certain event}) = 1$

iii. $P(\text{impossible event}) = 0$

iv. $0 \leq p \leq 1$

3. **Addition Theorem**

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$$

4. **Multiplication Theorem**

$$P(A \cap B) = P(A \text{ and } B) = P(A) \times P(B),$$

where $A$and $B$are independent events.

5. **Conditional Probability**

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \text{provided } P(B) \neq 0$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)}, \text{provided } P(A) \neq 0$$

6.

$$P(\bar{A}) = 1 - P(A)$$

CHAPTER- 12 (DERIVATIVES)

1) $\left(\frac{d}{dx}(x^n) = nx^{\,n-1}\right)$

2) $\left(\frac{d}{dx}(c) = 0,;\ c = \text{constant}\right)$

3) $\left(\frac{d}{dx}(\sin\ x) = \cos\ x\right)$

4) $\left(\frac{d}{dx}(\cos\ x) = -\sin\ x\right)$

5) $\left(\frac{d}{dx}(\tan\ x) = \sec^{\,2} x\right)$

6) $\left(\frac{d}{dx}(\cot\ x) = -\text{cosec}^{\,2} x\right)$

7) $\left(\frac{d}{dx}(\sec\ x) = \sec\ x \cdot \tan\ x\right)$

8) $\left(\frac{d}{dx}(\text{cosec}\ x) = -\text{cosec}\ x \cdot \cot\ x\right)$

9) $\left(\frac{d}{dx}(e^x) = e^x\right)$

10) $\left(\frac{d}{dx}(\log\ x) = \frac{1}{x}\right)$

11) $\left(\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}\right)$

12) $\left(\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}\right)$

13) $\left(\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}\right)$

14) $\left(\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}\right)$

15) $\left(\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}\right)$

16) $\left(\frac{d}{dx}(\text{cosec}^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}\right)$

Important formula

1.

$$\frac{d}{dx}(\sinh\ x) = \cosh\ x$$

2.

$$\frac{d}{dx}(\cosh\ x) = \sinh\ x$$

3.

$$\frac{d}{dx}(\tanh\ x) = \text{sech}^2 x$$

4.

$$\frac{d}{dx}(\coth\ x) = -\text{cosech}^2 x$$

5.

$$\frac{d}{dx}(\text{sech}\, x) = -\text{sech}\, x \cdot \tanh\ x$$

6.

$$\frac{d}{dx}(\text{cosech}\, x) = -\text{cosech}\, x \cdot \coth\ x$$

7.

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

8.

$$\frac{d}{dx}(\cosh^{-1} x) = -\frac{1}{\sqrt{x^2-1}}$$

9.

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$$

10.

$$\frac{d}{dx}(\coth^{-1} x) = -\frac{1}{x^2-1}$$

11.

$$\frac{d}{dx}(\text{\sech}^{-1}x) = \frac{1}{x\sqrt{1-x^2}}$$

12.

$$\frac{d}{dx}(\text{\cosech}^{-1}x) = -\frac{1}{x\sqrt{x^2+1}}$$

Chapter -13

APPLICATION OF DERIVATIVES

1. L'Hospital's Rule

If $f(x)$and $g(x)$and their derivatives $f'(x)$, $g'(x)$are continuous at $x = a$, and $f(a) = g(a) = 0$, then:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)}$$

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2. Rate Measures (Derivatives)

1. **Derivative of n^nw.r.t n:**

$$\frac{d}{dn}(x^n) = \mathrm{nx}^{n-1}$$

2. **Derivative of v^2w.r.t t(Chain Rule):**

$$\frac{dv^2}{dt} = \frac{dv^2}{dv} \cdot \frac{dv}{dt} = 2v\frac{dv}{dt}$$

3. **Derivative of πr^2w.r.t n(Chain Rule):**

$$\frac{d(\pi r^2)}{dx} = \frac{d(\pi r^2)}{dr} \cdot \frac{dr}{dx} = 2\pi r \cdot \frac{dr}{dx}$$

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3. Volume Formulas

- **Sphere:**

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3$$

- **Cylinder:**

$$V_{\text{cylinder}} = \pi r^2 h$$

- **Cone:**

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

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4. Surface Area

- **Sphere:**

$$S.A_{\text{sphere}} = 4\pi r^2$$

CHAPTER -14 (Antiderivative)

1.

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$$

2.

$$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\log\frac{a+x}{a-x} + c$$

3.

$$\int \frac{1}{x^2-a^2}\,dx = \frac{1}{2a}\log\frac{x-a}{x+a} + c$$

4.

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c$$

5.

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left(x+\sqrt{x^2+a^2}\right) + c$$

or

$$= \sinh^{-1}\frac{x}{a} + c$$

6.

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left(x+\sqrt{x^2-a^2}\right) + c$$

or

$$= \cosh^{-1}\frac{x}{a} + c$$

Chapter – 15

[Differential Equations]

1. Variable separated form: $[Y, dy = X, dx]$

2. Homogeneous form: $\left[\frac{dy}{dx} = f\left(\frac{y}{x}\right)\right]$

   $(x^2 = y^2)$

3. Linear form: $\left[\frac{dy}{dx} + Py = 0\right]$

4. Exact form: $[M(x, y), dx + N(x, y), dy = 0]$

5. Variable separated form: $[Y, dy = X, dx]$

COMPUTATIONAL METHODS

CHAPTER -16

SYSTEM OF LINEAR EQUATIONS

A) GAUSS ELIMINATION METHOD

→A system of linear equation is said to be consistent if it has either one solution or infinitely many solutions and system is said to **inconsistent** if it has no solution.

B) GAUSS-SEIDEL METHOD

FORMULAE AND IMPORTANT POINTS

1. **Diagonally Dominant**

A system

$$a_{11}X_1 + a_{12}X_2 + a_{13}X_3 = b_1$$
$$a_{21}X_1 + a_{22}X_2 + a_{23}X_3 = b_2$$
$$a_{31}X_1 + a_{32}X_2 + a_{33}X_3 = b_3$$

is diagonally dominant if

$$| a_{11} | > | a_{12} | + | a_{13} |$$
$$| a_{22} | > | a_{21} | + | a_{23} |$$
$$| a_{33} | > | a_{31} | + | a_{32} |$$

Chapter – 17

Linear Programming : (5 marks) ✓

1. **Slack variable**
   The non-negative variable by the addition of which on the left hand side of the inequality, changes the inequality into equality, is known as slack variable.

2. **Surplus variable**
   The non-negative variable by the subtraction of which on the left hand side of the inequality, changes the inequality into equality, is known as surplus variable.

16. STATICS

FORMULAE AND IMPORTANT POINTS

1. **Resultant of Two Forces Acting at A Point**

$$R^2 = P^2 + Q^2 + 2PQ\cos\ \alpha$$

$$\theta = \tan^{-1}\left(\frac{Q\sin\ \alpha}{P + Q\cos\ \alpha}\right)$$

2. **Resolution of a Given Force in Two Given Directions**

$$P = \frac{F\sin\ \beta}{\sin\ (\alpha + \beta)}, Q = \frac{F\sin\ \alpha}{\sin\ (\alpha + \beta)}$$

3. **Resolved Parts of a Number of Coplanar Concurrent Forces**

$$R\cos\ \theta = P_1\cos\ \alpha_1 + P_2\cos\ \alpha_2 + P_3\cos\ \alpha_3 + \cdots = X$$

$$R\sin\ \theta = P_1\sin\ \alpha_1 + P_2\sin\ \alpha_2 + P_3\sin\ \alpha_3 + \cdots = Y$$

$$R = \sqrt{X^2 + Y^2}$$

$$\theta = \tan^{-1}\frac{Y}{X}$$

17. DYNAMICS

A. NEWTON'S LAW OF MOTION

FORMULAE AND IMPORTANT POINTS

1.

Momentum $= mv$

Change in momentum $= mv - mu$

Average rate of change of momentum $= \dfrac{m(v-u)}{t}$

2.

Upward force: $R = m(a+g)$

Downward force: $R = m(g-a)$

3.

Impulsive force: $F = \dfrac{d(mv)}{dt}$,as m is constant

4. **Principle of conservation of linear momentum**

$$mv - MV = 0$$

$$m_1v_1 + m_2v_2 = (m_1 + m_2)V$$

B. PROJECTILES

FORMULAE AND IMPORTANT POINTS

1. **Equations of Motion of a Projectile**

- For the vertical motion:

$$V\sin\ \theta = u\sin\ \alpha - gt$$

- For the horizontal motion:

$$V\cos\ \theta = u\cos\ \alpha$$

2. **Time to reach the greatest height**

$$t = \frac{u\sin\ \alpha}{g}$$

3. **Time of Flight and Range**

$$T = \frac{2u\sin\ \alpha}{g}$$

$$R = \frac{u^2\sin\ 2\alpha}{g}$$

4. **Maximum horizontal range**

$$R_{\text{max}} = \frac{u^2}{g}$$

5. **Greatest height**

$$H = \frac{u^2\sin^2\alpha}{2g}$$